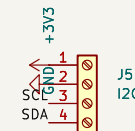
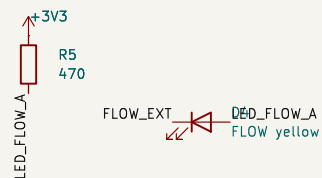


Left free: GPIO2 and GPIO8/9 (strapping; GPIO8 = module LED, GPIO9 = BOOT), GPIO21.

Crystal 32.768 kHz, 12.5 pF load.

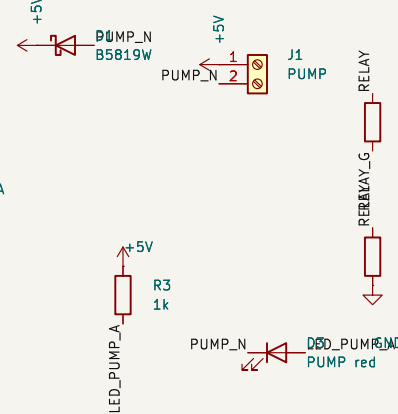


output, so it blinks with every pulse (no GPIO needed).

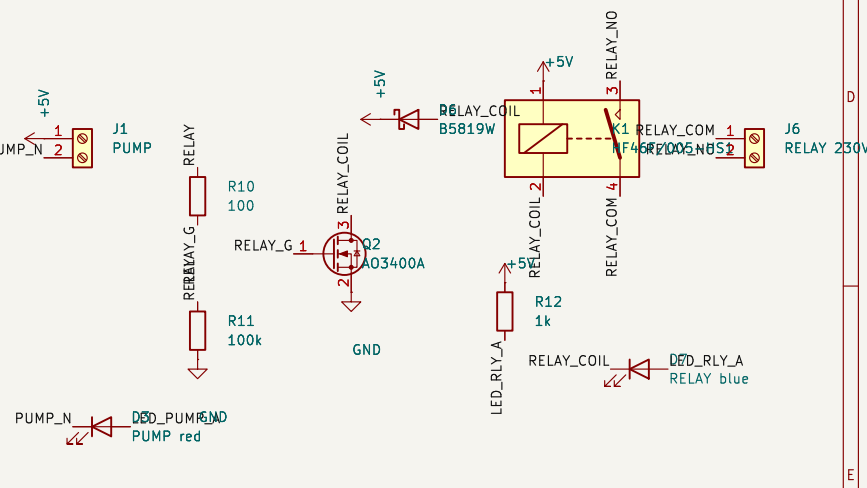


LED_STATUS_A
STATUS green

C1 rides out the start-up surge of the pump and the relay on USB 5 V.

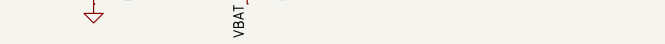


terminal; 230 V copper keeps ≥ 5 mm from everything else (.kicad_dru rule).



Mounting: 3 x 4.2 mm holes for 5 mm nylon reverse-mount snap standoffs, bases taped to the box floor (any box).



	
48 x 44 mm; 4 Mini+ 4 AHT20 sticks per 100 x 100 panel; 3 snap-standoff holes (any box)	
Generated by tools/gen_sch.py from tools/design.py – edit design.py and regenerate	
Agri Sensors and Controls	
Sheet: /	
File: ASC–MiniC3.kicad_sch	
Title: ASC Student Kit – Mini–C3 rev B (low cost, FG1 common)	
Size: A3	Date: 2026–10–10
KiCad E.D.A. 9.0.7	Rev: B Id: 1/1